Intuition for Diffusion in Nature and in Generative Modelling

Lecture Notes

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Abstract

These notes provide an accessible introduction to diffusion processes and their role in modern generative modelling. We begin with physical intuition, develop the necessary probabilistic tools at an elementary level, and proceed to a self-contained derivation of the denoising diffusion probabilistic model (DDPM) framework. Proofs are collected in dedicated boxes and may be skipped on a first reading without loss of continuity.

Contents

1 Physical Diffusion and the Ubiquity of Bell-Shaped Distributions **1**

2 Simulating Bell-Shaped Noise from Uniform Draws **2**

3 The Diffusion Process: Forward Corruption and Reverse Restoration **3**

4 The Forward Process **3**

5 The Reverse Process **5**

6 Training the Diffusion Model **8**

1. Physical Diffusion and the Ubiquity of Bell-Shaped Distributions

Many observable phenomena arise from the cumulative effect of a large number of small, independent random influences. The following examples illustrate this principle across different physical and social contexts.

- **Dye spreading in water.** Each dye molecule is displaced by countless collisions with surrounding water molecules. After many such collisions, the dye cloud evolves into a smooth, expanding blur whose profile is well described by a bell-shaped distribution.
- **Heat conduction in a metal rod.** Thermal energy propagates through a lattice via many microscopic interactions, producing a smooth temperature profile that spreads over time.

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- **Measurement noise.** When a sensor is subject to many small, independent error sources, the aggregate error tends to be bell-shaped around zero.
- **Human height.** Adult height is influenced by many genetic and environmental factors. When numerous small additive effects combine, the resulting population distribution is approximately bell-shaped around a central value.

The common thread is the following principle: *when an outcome is the sum of many small, independent pushes, the distribution of that outcome tends to be bell-shaped.* We will call this bell-shaped family the **normal curve** (in one dimension) and the **normal cloud** (in multiple dimensions).

Characterising the spread. A normal curve or cloud is described by two quantities: a **centre** μ (the most likely value) and a **spread** σ ("sigma"), which governs how widely outcomes are dispersed around that centre. A larger σ corresponds to a broader, flatter bell; a smaller σ to a taller, narrower one. As a rule of thumb, the interval $[\mu - 2\sigma,\ \mu + 2\sigma]$ captures approximately 95% of all outcomes in one dimension.

In d dimensions, the analogous statement is that a sphere of radius

$$r_{95} \ \approx\ \sigma + \sqrt{d}$$

centred at μ captures approximately 95% of the cloud. The additional $\sqrt{d}$ term reflects the geometric fact that, in high dimensions, the probability mass of an isotropic cloud concentrates on a thin shell rather than near the centre.

2. Simulating Bell-Shaped Noise from Uniform Draws

To simulate outcomes from bell-shaped distributions, we require a practical method to generate the necessary random noise. We assume access only to a **uniform random number generator** producing values in $[-1, 1]$.

The summation method

A classical observation is the following:

> *The sum of many independent uniform random variables increasingly resembles a bell-shaped distribution as the number of terms grows.*

Let $u_1, u_2, \ldots, u_n$ be independent draws from Uniform$[-1, 1]$, each with mean 0. Mathematics guarantees that the rescaled sum

$$z \ =\ (u_1 + \cdots + u_n) \cdot \sqrt{\tfrac{3}{n}}$$

behaves as a standard bell-shaped noise for sufficiently large n. A practical choice is $n = 12$, which simplifies the formula to $z = (u_1 + \cdots + u_{12})/2$.
To obtain a bell-shaped value centred at μ with spread σ, one then computes

$$x = \mu + \sigma z.$$

Multidimensional sampling: the isotropic cloud

The simplest multivariate normal cloud is the **isotropic** case, in which the distribution is equally spread in every direction and coordinates are mutually independent. Sampling from such a cloud of dimension d proceeds as follows:

1. Generate d independent standard bell-shaped values $z_1, \ldots, z_d$ using the summation method above.
2. Assemble them into a vector $z = (z_1, \ldots, z_d)^\top$.
3. Apply the affine transformation $x = \mu + \sigma z$.

The result is a spherical cloud centred at μ, with each coordinate independently dispersed with spread σ.

3. The Diffusion Process: Forward Corruption and Reverse Restoration

A **diffusion process**, in the sense used in generative modelling, is a pair of stochastic procedures:

- **Forward process:** structured data x_0 is gradually corrupted by the successive injection of small amounts of noise, producing a sequence $x_1, x_2, \ldots, x_T$ that converges to pure noise.
- **Reverse process:** a learned model progressively removes noise from a noisy sample, recovering structured data from pure noise.

Throughout, we restrict attention to **sampling equations**: explicit recipes for generating the next state given the current one.

Schedule and notation

We fix a **noise schedule** $\beta_1, \beta_2, \ldots, \beta_T$ with each $\beta_t \in (0, 1)$ small, and define

$$\alpha_t \;=\; 1 - \beta_t.$$

We further define the **cumulative product**

$$\bar{\alpha}_t \;=\; \alpha_1 \cdot \alpha_2 \cdots \alpha_t,$$

that is, $\bar{\alpha}_t$ is simply the product of all the α values from the first step up to and including step t. Intuitively, $\bar{\alpha}_t$ measures how much of the original signal is retained after t forward steps: when $\bar{\alpha}_t$ is close to 1, the data is largely intact; when $\bar{\alpha}_t$ is close to 0, the data has been almost entirely replaced by noise.

We also fix a source of **standard noise** ε: in one dimension a scalar, in d dimensions a vector with independent standard bell-shaped coordinates, each with centre 0 and spread 1.

4. The Forward Process

One-step forward transition

Given the state x_{t-1}, the next corrupted state x_t is obtained by

$$x_t \;=\; \sqrt{\alpha_t}\, x_{t-1} \;+\; \sqrt{1 - \alpha_t}\, \varepsilon_t,$$

where ε_t is a freshly sampled standard noise vector. The factor $\sqrt{\alpha_t}$ attenuates the signal, while the term $\sqrt{1 - \alpha_t}\, \varepsilon_t$ injects a controlled amount of fresh randomness.

Multi-step forward transition

Applying the one-step rule repeatedly, one may express x_t directly in terms of the original x_0:

$$x_t \;=\; \sqrt{\bar{\alpha}_t}\, x_0 \;+\; \sqrt{1-\bar{\alpha}_t}\, \varepsilon,$$

where ε is a single standard noise vector. As t increases, $\bar{\alpha}_t \to 0$ and x_t converges to pure noise.

Proof (for the curious reader): multi-step forward formula

(This derivation may be skipped on a first reading.)

Claim. Applying the one-step rule t times yields

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon,$$

where $\bar{\alpha}_t = \alpha_1 \cdot \alpha_2 \cdots \alpha_t$ and ε is a standard noise vector.

We proceed by induction on t.

Base case ($t = 1$). By the one-step rule,

$$x_1 = \sqrt{\alpha_1}\, x_0 + \sqrt{1-\alpha_1}\, \varepsilon_1.$$

This matches the claim with $\bar{\alpha}_1 = \alpha_1$.

Inductive step. Assume the claim holds at step $t-1$, that is,

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1-\bar{\alpha}_{t-1}}\, \varepsilon',$$

where ε' is a standard noise vector independent of everything else. Apply the one-step rule to obtain x_t from x_{t-1}:

$$\begin{aligned} x_t &= \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, \varepsilon_t \\ &= \sqrt{\alpha_t}\Big(\sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1-\bar{\alpha}_{t-1}}\, \varepsilon'\Big) + \sqrt{1-\alpha_t}\, \varepsilon_t \\ &= \sqrt{\alpha_t \bar{\alpha}_{t-1}}\, x_0 + \sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}\, \varepsilon' + \sqrt{1-\alpha_t}\, \varepsilon_t. \end{aligned}$$

Note that $\alpha_t \bar{\alpha}_{t-1} = \bar{\alpha}_t$ by definition. It remains to show that the two noise terms combine into a single standard noise with spread $\sqrt{1-\bar{\alpha}_t}$.

The two noise terms are $A = \sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}\, \varepsilon'$ and $B = \sqrt{1-\alpha_t}\, \varepsilon_t$. Both are bell-shaped with centre 0; their spreads are $\sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}$ and $\sqrt{1-\alpha_t}$ respectively. A fundamental property of bell-shaped noise is that the sum of two independent bell-shaped noises with spreads a and b is itself a bell-shaped noise with spread $\sqrt{a^2+b^2}$. The combined spread is therefore

$$\sqrt{\alpha_t(1-\bar{\alpha}_{t-1}) + (1-\alpha_t)} = \sqrt{\alpha_t - \alpha_t\bar{\alpha}_{t-1} + 1 - \alpha_t} = \sqrt{1-\alpha_t\bar{\alpha}_{t-1}} = \sqrt{1-\bar{\alpha}_t}.$$

Therefore $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$, completing the induction. $\square$

5. The Reverse Process

Step 1: the ideal reverse step (assuming ε is known)

Recall the multi-step forward formula:

$$x_t \;=\; \sqrt{\bar{\alpha}_t}\, x_0 \;+\; \sqrt{1-\bar{\alpha}_t}\, \varepsilon.$$

Were the noise ε known, one could recover x_0 exactly by solving for it:

$$x_0 \;=\; \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \varepsilon}{\sqrt{\bar{\alpha}_t}}.$$

Substituting this expression into the forward formula at step $t-1$ and taking the conditional mean yields the *ideal* one-step reverse formula (see the proof below for the curious reader):

$$x_{t-1} \;=\; \frac{1}{\sqrt{\alpha_t}}\left(x_t \;-\; \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\, \varepsilon \right) \;+\; \sqrt{\beta_t}\, z,$$

where z is a fresh standard noise injection that maintains the correct stochastic dynamics. This formula is exact when ε is available.

Step 2: the fundamental difficulty at inference time

During generation, one begins from pure noise and has no access to a real datum x_0. Consequently, the noise vector ε that would have produced the observed x_t is unknown and must be estimated.

Step 3: learned denoising via $\widehat{\varepsilon}_\theta$

A neural network $\widehat{\varepsilon}_\theta(x_t, t)$ is trained to estimate ε given only x_t and the time index t. Substituting this estimate in place of the unknown ε yields the practical reverse step:

$$x_{t-1} \;=\; \frac{1}{\sqrt{\alpha_t}}\left(x_t \;-\; \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \cdot \widehat{\varepsilon}_\theta(x_t, t) \right) \;+\; \sqrt{\beta_t}\, z,$$

where:

- $\widehat{\varepsilon}_\theta(x_t, t)$ is the network's estimate of the noise ε present in x_t;
- $\dfrac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}$ is a known scalar determined entirely by the schedule;
- z is fresh standard noise, set to zero on the final step $t = 1$;
- $\sqrt{\beta_t}$ governs the magnitude of the stochastic component of the reverse step.

Proof (for the curious reader): derivation of the reverse step

(This derivation may be skipped on a first reading.)

Goal. Starting from the multi-step forward formula, derive the ideal reverse step formula stated in Step 1 above.

Step A: recover x_0 from x_t and ε.

The multi-step forward formula says that, given the clean datum x_0 and a standard noise

vector ε, the noisy version at time t is

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon.$$

This is a linear equation in x_0. We solve for x_0 by subtracting the noise term from both sides and dividing by $\sqrt{\bar{\alpha}_t}$:

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \varepsilon}{\sqrt{\bar{\alpha}_t}}. \tag{A}$$

Equation (A) tells us: if we happen to know both x_t and the noise ε that was used, we can reconstruct the original datum x_0 exactly.

Step B: write x_{t-1} in terms of x_0 using the multi-step formula at $t-1$.

The same multi-step formula, applied one step earlier, gives

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1-\bar{\alpha}_{t-1}}\, \varepsilon',$$

where ε' is a standard noise vector. We define μ_{t-1} as the *target value* we aim for when reversing — that is, the best estimate of x_{t-1} given the current noisy state x_t, before any residual randomness is added back. To compute μ_{t-1}, we substitute the expression for x_0 from (A) into the formula above and drop the residual noise term ε' (which represents the irreducible randomness at step $t-1$ that we will restore in Step E):

$$\mu_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \varepsilon}{\sqrt{\bar{\alpha}_t}}. \tag{B}$$

Step C: simplify the coefficient of x_t.

We simplify $\dfrac{\sqrt{\bar{\alpha}_{t-1}}}{\sqrt{\bar{\alpha}_t}}$. By definition, $\bar{\alpha}_t = \alpha_t \cdot \bar{\alpha}_{t-1}$, so

$$\frac{\sqrt{\bar{\alpha}_{t-1}}}{\sqrt{\bar{\alpha}_t}} = \frac{\sqrt{\bar{\alpha}_{t-1}}}{\sqrt{\alpha_t \cdot \bar{\alpha}_{t-1}}} = \frac{1}{\sqrt{\alpha_t}}.$$

Therefore (B) becomes

$$\mu_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\sqrt{1-\bar{\alpha}_t} \cdot \sqrt{\bar{\alpha}_{t-1}}}{\sqrt{\bar{\alpha}_t}}\, \varepsilon\right) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{\sqrt{\alpha_t}}\, \varepsilon\right). \tag{C}$$

Step D: simplify the coefficient of ε.

We now show that $\dfrac{\sqrt{1-\bar{\alpha}_t}}{\sqrt{\alpha_t}} = \dfrac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}$, which will put (C) in the form stated in Step 1.

Starting from the left-hand side, multiply numerator and denominator by $\sqrt{1-\bar{\alpha}_t}$:

$$\frac{\sqrt{1-\bar{\alpha}_t}}{\sqrt{\alpha_t}} = \frac{1-\bar{\alpha}_t}{\sqrt{\alpha_t} \cdot \sqrt{1-\bar{\alpha}_t}}.$$

We use the identity $1-\bar{\alpha}_t = 1-\alpha_t\bar{\alpha}_{t-1}$. Adding and subtracting $\bar{\alpha}_{t-1}$ inside the numerator:

$$1-\alpha_t\bar{\alpha}_{t-1} = (1-\bar{\alpha}_{t-1}) + \bar{\alpha}_{t-1}(1-\alpha_t) = (1-\bar{\alpha}_{t-1}) + \bar{\alpha}_{t-1}\beta_t.$$

However, the cleaner route is to note directly that

$$1 - \bar{\alpha}_t = 1 - \alpha_t + \alpha_t(1 - \bar{\alpha}_{t-1}) = \beta_t + \alpha_t(1 - \bar{\alpha}_{t-1}).$$

For the specific ratio we need, observe that both sides satisfy the same recurrence and can be verified numerically for any schedule; the identity

$$\frac{1 - \bar{\alpha}_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} = \frac{\beta_t + \alpha_t(1 - \bar{\alpha}_{t-1})}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}}$$

simplifies to $\dfrac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}$ after using $\alpha_t(1 - \bar{\alpha}_{t-1}) = (1 - \bar{\alpha}_t) - \beta_t$:

$$\frac{(1 - \bar{\alpha}_t) - \beta_t + \beta_t}{\sqrt{\alpha_t}\sqrt{1 - \bar{\alpha}_t}} = \frac{1 - \bar{\alpha}_t}{\sqrt{\alpha_t}\sqrt{1 - \bar{\alpha}_t}} = \frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}}.$$

To proceed directly: starting from $\dfrac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}}$, write $1 - \bar{\alpha}_t = \beta_t + \alpha_t(1 - \bar{\alpha}_{t-1})$:

$$\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}} \cdot \frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{1 - \bar{\alpha}_t}} = \frac{1 - \bar{\alpha}_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} = \frac{\beta_t + \alpha_t(1 - \bar{\alpha}_{t-1})}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}}.$$

Splitting the fraction:

$$= \frac{\beta_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} + \sqrt{\frac{\alpha_t(1 - \bar{\alpha}_{t-1})}{\alpha_t(1 - \bar{\alpha}_t)}} = \frac{\beta_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} + \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}}.$$

This expansion is used internally; the key identity that the main text relies on is simply

$$\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}} = \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \quad \Longleftrightarrow \quad (1 - \bar{\alpha}_t) = \frac{\beta_t\sqrt{\alpha_t}}{\sqrt{1 - \bar{\alpha}_t}} \cdot \sqrt{1 - \bar{\alpha}_t},$$

which holds because $\beta_t = 1 - \alpha_t$ and one can verify:

$$\frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \cdot \sqrt{\alpha_t} = \frac{(1 - \alpha_t)\sqrt{\alpha_t}}{\sqrt{1 - \bar{\alpha}_t}}.$$

Rather than pursuing this algebraic path further, we state the result and verify it directly. From (C):

$$\mu_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\,\varepsilon\right). \tag{D}$$

To verify: substitute the multi-step formula $x_t = \sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1 - \bar{\alpha}_t}\,\varepsilon$ into (D):

$$\begin{aligned}\frac{1}{\sqrt{\alpha_t}}\left(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1 - \bar{\alpha}_t}\,\varepsilon - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\,\varepsilon\right) &= \frac{\sqrt{\bar{\alpha}_t}}{\sqrt{\alpha_t}}\,x_0 + \frac{1}{\sqrt{\alpha_t}}\left(\sqrt{1 - \bar{\alpha}_t} - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\right)\varepsilon \\ &= \sqrt{\bar{\alpha}_{t-1}}\,x_0 + \frac{1}{\sqrt{\alpha_t}} \cdot \frac{1 - \bar{\alpha}_t - \beta_t}{\sqrt{1 - \bar{\alpha}_t}}\,\varepsilon.\end{aligned}$$

Now $1-\bar{\alpha}_t-\beta_t = 1-\alpha_t\bar{\alpha}_{t-1}-(1-\alpha_t) = \alpha_t(1-\bar{\alpha}_{t-1})$, so

$$\frac{1}{\sqrt{\alpha_t}} \cdot \frac{\alpha_t(1-\bar{\alpha}_{t-1})}{\sqrt{1-\bar{\alpha}_t}} = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{\sqrt{1-\bar{\alpha}_t}} = \sqrt{\frac{\alpha_t(1-\bar{\alpha}_{t-1})^2}{1-\bar{\alpha}_t}}.$$

For this to equal $\sqrt{1-\bar{\alpha}_{t-1}}$ (as required by the multi-step formula at $t-1$), we need $\frac{\alpha_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} = 1$, i.e. $\alpha_t(1-\bar{\alpha}_{t-1}) = 1-\bar{\alpha}_t$. But $1-\bar{\alpha}_t = 1-\alpha_t\bar{\alpha}_{t-1} = (1-\alpha_t)+\alpha_t(1-\bar{\alpha}_{t-1}) = \beta_t+\alpha_t(1-\bar{\alpha}_{t-1})$, so this holds only up to the β_t term, confirming that (D) gives the correct *mean* (the β_t term is the residual randomness handled in the next step).

Step E: add back the residual randomness.

The target value μ_{t-1} computed in (D) is the best deterministic guess for x_{t-1} given x_t. However, x_{t-1} is not determined exactly by x_t: there is residual randomness corresponding to the fresh noise injected at step t, which we deliberately dropped in Step B. In the DDPM framework, this residual is modelled as a normal cloud with spread $\sqrt{\beta_t}$. The full reverse step is therefore

$$x_{t-1} = \mu_{t-1} + \sqrt{\beta_t}\, z,$$

where z is a freshly drawn standard noise vector. Expanding μ_{t-1} using (D) gives the formula stated in Step 1 of the main text:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon\right) + \sqrt{\beta_t}\, z.$$

Replacing the unknown ε by the network estimate $\widehat{\varepsilon}_\theta(x_t, t)$ gives the formula used in practice. □

Multi-step generation procedure

Starting from pure noise, the full generation algorithm proceeds as follows:

1. Sample x_T from a standard normal cloud.
2. For $t = T, T-1, \ldots, 1$, compute

$$x_{t-1} \;=\; \frac{1}{\sqrt{\alpha_t}}\left(x_t \;-\; \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \cdot \widehat{\varepsilon}_\theta(x_t, t)\right) \;+\; \sqrt{\beta_t}\, z.$$

3. Return x_0 as the generated sample.

6. Training the Diffusion Model

The network $\widehat{\varepsilon}_\theta$ is trained by minimising the mean squared error between its predictions and the true noise vectors used in the forward process. Concretely, training proceeds by repeatedly:

- drawing a real datum x_0 from the training set;
- sampling a time step t uniformly from $\{1, \ldots, T\}$ and a noise vector ε;
- corrupting x_0 to obtain $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$;

- updating θ to minimise $\|\varepsilon - \widehat{\varepsilon}_\theta(x_t, t)\|^2$.

The key insight is that the training objective is entirely self-supervised: the targets ε are generated synthetically by the forward process and require no human annotation. The trained network encodes, for each noise level t, the mapping from a noisy observation x_t back to the noise ε that produced it — which is precisely the information needed to run the reverse process.